9th Class • Punjab Board • PCTB New Syllabus

Chapter 6: Biomolecules

Solved notes in book order — MCQs, short answers, detailed answers, inquisitive questions and key points.

Chapter 6 • Units 6.1 – 6.6 • Exercise solved in book order

What this chapter is about

• Biomolecules are the molecules produced by living organisms, i.e. carbohydrates, proteins, lipids and nucleic acids. They make up 85–90% of the dry mass of a cell, and the remaining 1–1.5% is made of inorganic salts.

• Carbohydrates are the primary source of energy; proteins are the building blocks and the working molecules; lipids store energy and make the cell membrane; nucleic acids (DNA and RNA) store and carry the hereditary information.

• The last section explains how the information of DNA is copied into mRNA (transcription) and then into proteins (translation) — the Central Dogma.

• The exercise (11 MCQs, 9 short answers, 6 detailed answers and 2 inquisitive questions) is solved here in book order.

A. Multiple Choice Questions

Tick the correct option. The reason after each answer tells you why the other options are wrong.

1. What is the primary function of carbohydrates?

a) Provide energy
b) Act as enzymes
c) Regulate processes
d) Make membranes

Answer: a) Provide energy — Carbohydrates are broken down in respiration and give energy (glucose is the body's main fuel). Enzymes and regulators are proteins, and membranes are mainly made of lipids and proteins.

2. How will you differentiate between monosaccharides and polysaccharides?

a) Monosaccharides are single sugars
b) Polysaccharides are sweet in taste
c) Monosaccharides are present in plant cell wall
d) Polysaccharides dissolve easily

Answer: a) Monosaccharides are single sugars — Monosaccharides are single sugar molecules (glucose, fructose) and are sweet and soluble; polysaccharides are large molecules of many sugars, tasteless and insoluble.

3. What is true about cellulose?

a) It is sweet in taste
b) It is digestible by human digestive system
c) It provides structural support in plants
d) It is soluble in water

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Answer: c) It provides structural support in plants — Cellulose is a polysaccharide of glucose units that forms the plant cell wall, so it gives support. It is not sweet, not soluble in water, and humans cannot digest it (it is roughage).

4. Which of the following proteins is involved in oxygen transport?

a) Insulin
b) Haemoglobin
c) Collagen
d) Keratin

Answer: b) Haemoglobin — Haemoglobin in the red blood cells carries oxygen from the lungs to the tissues. Insulin is a hormone, collagen is a structural protein and keratin is present in hair and nails.

5. Which component of an amino acid determines its unique properties?

a) Amino group
b) Carboxyl group
c) R group (side group)
d) Hydrogen group

Answer: c) R group (side group) — All amino acids have the same amino group and carboxyl group; the R group is different in each amino acid, so it decides the properties of that amino acid.

6. Which proteins are involved in defence against pathogens?

a) Antibodies
b) Myosin
c) Fibrinogen
d) Haemoglobin

Answer: a) Antibodies — Antibodies are proteins which recognise and destroy pathogens. Myosin is for muscle contraction, fibrinogen for blood clotting and haemoglobin for oxygen transport.

7. Which of the following are the units of most lipids?

a) Amino acids
b) Fatty acids and glycerol
c) Nucleotides
d) Simple sugars

Answer: b) Fatty acids and glycerol — Most lipids are triglycerides: one glycerol molecule joined to three fatty acids. Amino acids are units of proteins, nucleotides of nucleic acids and simple sugars of carbohydrates.

8. How do unsaturated fatty acids differ from saturated fatty acids?

a) They have more hydrogen atoms
b) They contain double bonds in their hydrocarbon chains
c) They are solid at room temperature
d) They are found only in animal fats

Answer: b) They contain double bonds in their hydrocarbon chains — Unsaturated fatty acids have one or more double bonds between carbon atoms, which is why they are liquid at room temperature; saturated fatty acids have only single bonds and are solid.

9. Which of the following is NOT a function of proteins?

a) Transport oxygen in the blood
b) Carry genetic information
c) Help in digesting food
d) Fight against pathogens

Answer: b) Carry genetic information — Genetic information is carried by nucleic acids (DNA and RNA). Proteins transport oxygen, act as enzymes (digestion)

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and as antibodies (defence).

10. Which components make up a nucleotide?

a) Sugar, phosphate, nitrogenous base
b) Amino acid, sugar, nitrogenous base
c) Fatty acid, phosphate, nitrogenous base
d) Protein, sugar, nitrogenous base

Answer: a) Sugar, phosphate, nitrogenous base — A nucleotide is made of three parts: a pentose sugar (ribose or deoxyribose), a phosphate group and one nitrogenous base.

11. Which nitrogenous base is found in RNA but not in DNA?

a) Adenine
b) Thymine
c) Uracil
d) Guanine

Answer: c) Uracil — RNA has uracil (U) in place of the thymine (T) of DNA; adenine and guanine are present in both.

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B. Write Short Answers

1. What are the main functions of carbohydrates in the body?

1. They are the primary source of energy: glucose is used by the cells in respiration and gives energy for the activities of the body.

2. They spare the proteins: when enough carbohydrates are available, the body does not break down its proteins for energy.

3. Dietary fibre (cellulose) helps in the proper bowel movements and prevents constipation.

4. They help in the proper functioning of joints and are used in the synthesis of glycoproteins and glycolipids.

5. Plants store carbohydrates as starch and animals as glycogen — these are carbohydrates between meals, and cellulose forms the cell wall of plants.

2. How do the three groups of carbohydrates differ in taste?

Monosaccharides (glucose, fructose, galactose) are single sugar molecules; they are easily soluble in water and are **sweet** in taste. Fructose is the sweetest of them.

Disaccharides (sucrose, lactose, maltose) are made of two monosaccharides; they are **sweet** too — sucrose (table sugar) is very sweet — but they dissolve less easily than monosaccharides.

Polysaccharides (starch, glycogen, cellulose) are made of hundreds to thousands of glucose units and are the most abundant carbohydrates; they are generally **insoluble in water and tasteless** (not sweet).

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3. Name two common monosaccharides and two disaccharides.

Monosaccharides: glucose (grape sugar) and fructose (fruit sugar). (Galactose is another.)

Disaccharides: sucrose (table sugar) and lactose (milk sugar). (Maltose is another.)

4. Which monosaccharides make a sucrose molecule?

Sucrose is made of two monosaccharides: one molecule of **glucose** and one molecule of **fructose**, joined together with the loss of a water molecule. It is found in sugarcane, sugar beet and honey, and it is the sugar we use in the home.

5. Give an example of a storage polysaccharide in plants.

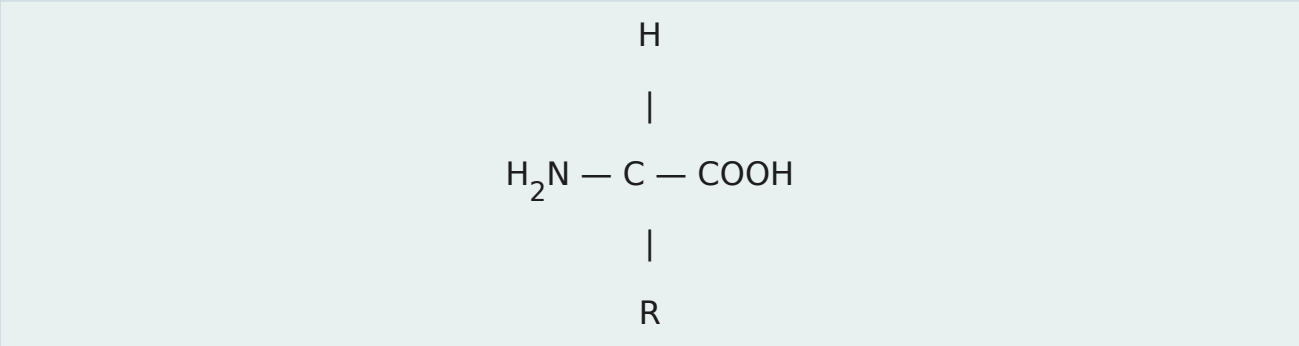
Answer: Starch

Starch is a storage polysaccharide formed in plants, e.g. in potato, wheat, rice and maize. It is made of hundreds to thousands of glucose units joined in straight and branched chains, and it is stored in the roots, stems and seeds of plants. Animals eat starch and break it down into glucose during digestion.

6. Define amino acid and draw its structure.

Definition: an amino acid is a small molecule made of an amino group, a carboxyl group and a side group which is attached to a carbon atom. Amino acids are the units of proteins — about 20 different types of amino acids are found in the proteins of living organisms. When many amino acids are joined by peptide bonds in a chain, a protein is formed.

Structure of an amino acid (a general formula — draw the central carbon with four groups attached):



Where H_2N = amino group, COOH = carboxyl group and R = side group (different in each amino acid). The central carbon atom carries all four groups.

7. What are the basic components of lipids? Draw their structures.

Components: most lipids are made of two components:

1. Glycerol: an alcohol with three carbon atoms, and each carbon atom has a hydroxyl (OH) group attached to it.

2. Fatty acids: long hydrocarbon chains with a carboxyl (COOH) group at one end. They may be saturated (only single bonds between carbons, solid at room temperature) or unsaturated (one or more double bonds, liquid at room temperature).

One glycerol molecule joins with three fatty acids by ester bonds to make a fat or oil (triglyceride). Phospholipids contain a phosphate group and make the cell membrane.

Structure (write these two in the paper):

Glycerol: $CH_2OH — CHOH — CH_2OH$ Fatty acid: $CH_3—(CH_2)_n—COOH$

Fat = 1 glycerol + 3 fatty acids + 3 H_2O

8. What are the types of RNA? Write their functions.

There are three types of RNA:

1. Messenger RNA (mRNA): it carries the genetic message in the form of a nucleotide sequence from DNA to the ribosomes, and it decides the sequence of amino acids in the protein.

2. Transfer RNA (tRNA): it transfers specific amino acids to the ribosome so that they can be added in the correct sequence during protein synthesis.

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3. Ribosomal RNA (rRNA): it is a structural component of the ribosomes and helps them to work as the cellular machinery for protein synthesis.

9. Briefly describe the function of DNA.

DNA (deoxyribonucleic acid) is found in the chromosomes of the nucleus and contains the hereditary information. Its functions are:

1. it stores the hereditary information in the sequence of its nucleotides — this sequence decides the order of amino acids in a protein and so controls the characters and activities of the cell;

2. it copies itself (replication) before cell division, so that every daughter cell gets the same information; and

3. it transfers its information to mRNA (transcription), which then makes proteins on the ribosome. In short, DNA is the master molecule which controls and transfers the genetic information to the next generation.

Remember

• Carbohydrates → mono-, di- and polysaccharides (glucose, sucrose, starch, glycogen, cellulose).

• Proteins → amino acids (about 20 types) joined by peptide bonds.

• Lipids → glycerol + fatty acids (saturated or unsaturated).

• Nucleic acids → nucleotides (sugar + phosphate + nitrogenous base); DNA double-stranded, RNA single-stranded.

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C. Write Answers in Detail

1. Write a comprehensive note on the structures and roles of the three classes of carbohydrates.

Introduction: carbohydrates are organic compounds composed of carbon, hydrogen and oxygen, usually in the ratio 1:2:1. The word carbohydrate means "hydrated carbon". They are the primary source of energy for the body and are the most abundant biomolecules. They are classified into three classes.

1. Monosaccharides — structure: these are single sugar molecules made of 3 to 7 carbon atoms. They are easily soluble in water and sweet in taste; they cannot be broken down further by hydrolysis. Their examples are ribose (5C), glucose, fructose and galactose (6C each).

Monosaccharides — roles: glucose is the main fuel of the cell and is used in respiration to give energy; fructose is the sugar of fruits and honey; ribose is a part of nucleic acids and ATP. Monosaccharides are also the building units of di- and polysaccharides.

2. Disaccharides — structure: these are formed by the joining of two monosaccharide units, and they are less soluble and sweet. Their general formula is $C_{12}H_{22}O_{11}$. Examples: sucrose (glucose + fructose — table sugar, from sugarcane and sugar beet), lactose (glucose + galactose — milk sugar) and maltose (glucose + glucose — malt).

Disaccharides — roles: they are the form in which sugar is transported in many plants (sucrose) and the sugar of milk (lactose); they provide quick energy and are used in food industries, e.g. maltose in the production of beer and bread.

3. Polysaccharides — structure: these are large molecules composed of hundreds to thousands of glucose units joined in long chains; they are not soluble in water and are tasteless. Important ones are starch, glycogen and cellulose.

Polysaccharides — roles:

(a) Starch: a storage polysaccharide of plants found in potato, wheat, rice and maize; it is broken down into glucose during digestion.

(b) Glycogen: the animal starch, made of highly branched chains of glucose units, stored in the liver and muscles; it is broken down between meals to keep the blood glucose normal.

(c) Cellulose: a polysaccharide of glucose which forms the cell wall of plants; it is the most abundant carbohydrate in the world and gives structural support to plant cells. It cannot be digested by humans but works as roughage (dietary fibre) and helps the bowel movements.

Conclusion: monosaccharides give ready energy, disaccharides are the sugar of food and milk, and polysaccharides are the stores of energy (starch, glycogen) and the structural material (cellulose). The body digests all of them into glucose before using them.

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2. Discuss the functions of proteins.

Introduction: proteins are the most abundant biomolecules after carbohydrates; they are the building blocks of the cells and are the working molecules that perform a huge variety of jobs.

1. Growth and repair: proteins are an important part of cell membranes and of all tissues; they are needed for the growth of the body and for the repair of worn-out cells.

2. Structural role: collagen and keratin make almost all the structural parts of the body — collagen in skin, bone and connective tissue, keratin in hair, nails and the outer layer of skin.

3. Enzymes: enzymes that catalyse all the chemical reactions of the body, including the digestion of food, are proteins.

4. Hormones: many hormones, such as insulin, are proteins that regulate the body's metabolism, growth and blood glucose.

5. Movement: actin and myosin in the muscle cells are proteins that make the contraction of muscles; they also help the movement of cells and chromosomes.

6. Blood clotting: fibrinogen is a protein that makes the blood clot and prevents the loss of blood after an injury.

7. Defence: antibodies are proteins which fight the pathogens (disease-causing organisms) and give immunity.

8. Transport: haemoglobin carries oxygen in the blood, and many carrier proteins in the cell membrane move substances into and out of the cell.

9. Storage: casein of milk and ovalbumin of egg white store protein for the young one; other proteins also store iron and other nutrients.

Conclusion: proteins are called the "workhorses of the cell" because they not only build the body but also run, control and protect it.

3. What are fatty acids? Explain their types.

Definition: fatty acids are long hydrocarbon chains having a carboxyl (COOH) group at one end. They are the building units of lipids; three fatty acid molecules join one glycerol molecule to make a fat or oil.

Types of fatty acids:

1. Saturated fatty acids: they have only single bonds between their carbon atoms and carry the maximum number of hydrogen atoms (they are "saturated" with hydrogen). Because of straight chains they pack closely, so they are solid at room temperature. They are found in animal fats such as butter, ghee and lard, and are associated with an increased risk of heart disease.

2. Unsaturated fatty acids: they have one or more double bonds between their carbon atoms, so they carry fewer hydrogen atoms. A double bond makes a bend (kink) in the chain, so the molecules do not pack closely and they are liquid at room temperature. They are found in vegetable oils such as olive, sunflower and corn oil, and in fish. They are of two kinds: monounsaturated (one double bond) and polyunsaturated (more than one double bond).

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Saturated fatty acids	Unsaturated fatty acids
Only single bonds between carbon atoms.	One or more double bonds between carbon atoms.
Have the maximum number of hydrogen atoms.	Have fewer hydrogen atoms.
Solid at room temperature.	Liquid at room temperature.
Mostly found in animal fats (butter, ghee, lard).	Mostly found in vegetable oils and fish oil.
More harmful for the heart in excess.	Generally healthier; contain essential fatty acids.

4. Describe the basic structure of fats and oils. Explain how lipids play role in energy storage.

Basic structure: fats and oils are also called triglycerides. One molecule of glycerol joins with three molecules of fatty acids with the loss of three molecules of water, and a triglyceride is formed. If the fatty acids are saturated, the compound is solid and is called a fat (e.g. ghee); if they are unsaturated, it is liquid and is called an oil (e.g. sunflower oil). Fats and oils are esters of glycerol and fatty acids.

$$\text{Glycerol} + 3\ \text{fatty acids} \rightarrow \text{fat (triglyceride)} + 3\ H_2O$$

Role in energy storage:

1. Rich source of energy: lipids are a long-term energy reserve in the form of fats in adipose tissue. One gram of fat gives about twice the energy of one gram of carbohydrate, so the body stores its extra energy as fat.

2. Compact storage: fat is stored without water and in a compact form, so a large amount of energy is stored in a small mass; this is important for animals and for seeds such as sunflower and coconut.

3. Insulation: the fat under the skin (blubber) reduces the loss of heat, so it helps in temperature regulation; it also protects the internal organs as a cushion.

4. Reserve for hard times: when food is not available, the stored fat is broken down into fatty acids and glycerol and used as a fuel, so the animal can survive.

5. Other uses: lipids are essential components of cell membranes (phospholipids), they are the source of steroid hormones, and they help in the absorption of fat-soluble vitamins (A, D, E and K).

Conclusion: because fat stores the maximum energy in the minimum mass, it is the best long-term fuel store of the body; carbohydrates are used first and the lipids are kept as a reserve for later.

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5. Explain the double helix structure of DNA, and discuss the base pairing in this structure.

Introduction: DNA (deoxyribonucleic acid) is made of nucleotides; each nucleotide consists of a deoxyribose sugar, a phosphate group and one nitrogenous base. In 1953 James Watson and Francis Crick proposed the double helix model of DNA, using the X-ray diffraction studies of Rosalind Franklin and Maurice Wilkins. It is a double-stranded molecule.

Structure of the double helix:

1. The two strands of DNA run side by side and coil round each other in the form of a double helix, like a twisted ladder.

2. The two strands are held together by hydrogen bonds between the nitrogenous bases of the opposite strand.

3. The sugar and phosphate groups form the backbone (the sides of the ladder) and the bases point inward (the rungs of the ladder).

4. The two strands are antiparallel and complementary to each other, so the sequence of one strand decides the sequence of the other.

Base pairing: there are two types of nitrogenous bases in DNA — **purines** (adenine A and guanine G) and **pyrimidines** (thymine T and cytosine C). The pairing of the bases is specific: adenine always pairs with thymine and guanine always pairs with cytosine. There are two hydrogen bonds between A and T, and three hydrogen bonds between G and C. Because of this specific pairing, the sequence of one strand can be used to rebuild the other strand during replication. The sequence of nucleotides is the hereditary information of the cell.

6. Explain how the information in DNA is converted to RNA and then to proteins.

Introduction: the information of DNA is stored in the sequence of its nucleotides. It is expressed in two steps: transcription (DNA → RNA) and translation (RNA → protein). This flow of information is called the Central Dogma.

1. Transcription: the DNA molecule is a double helix, so its information is first copied into a single-stranded messenger RNA. In the nucleus the two strands of DNA separate, and one strand works as a template. The copying of a specific sequence of DNA nucleotides in the form of messenger RNA (mRNA) is called transcription. In RNA, uracil takes the place of thymine.

2. Processing and export: the mRNA carries the message out of the nucleus through the nuclear pores into the cytoplasm, where the ribosomes are present.

3. Translation: the ribosome reads the nucleotide sequence of mRNA in triplets (codons) and joins the specific amino acids according to it to form a protein. This process is known as translation. The transfer RNA (tRNA) brings the correct amino acids to the ribosome, and the ribosomal RNA (rRNA) of the ribosome joins them by peptide bonds.

4. The result: the sequence of nucleotides of DNA therefore decides the sequence of amino acids of the protein, and the sequence of amino acids decides the shape and function of the protein — and hence the character of the organism.

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Conclusion: DNA → mRNA → protein. In this way the hereditary information controls every activity of the cell.

D. Inquisitive Questions

1. Evaluate the importance of water in the functioning of biomolecules.

1. Universal solvent: water is the medium in which all the biomolecules are dissolved and all the reactions of the cell take place. Sugars, amino acids, salts and gases dissolve in it, so they can be transported and can meet each other.

2. Medium of chemical reactions: hydrolysis (breaking of a bond by water) and condensation (joining of two molecules with the loss of water) are the reactions by which carbohydrates, proteins and fats are built and broken. Digestion is nothing but hydrolysis of biomolecules by water with the help of enzymes.

3. Part of the biomolecules: water takes part in the formation of every biomolecule — a water molecule is lost whenever two amino acids, two sugars or a glycerol and fatty acid are joined. Water also forms hydrogen bonds with proteins and holds their shape.

4. Transport: water (as blood plasma, cell sap and lymph) carries nutrients, hormones and wastes throughout the body and inside the plant.

5. Temperature control: water has a high heat capacity and a high latent heat of vaporization, so it absorbs heat without a big change in its own temperature and cools the body by sweating and the plants by transpiration.

6. Structure and support: the turgor pressure of water makes plant cells firm; water is also the main part of cytoplasm (70–90%) and of all body fluids.

7. Protection: water lubricates the joints, the eyes and the alimentary canal, and it removes the wastes through urine.

Conclusion: water is called the "universal solvent" and the "medium of life": no biomolecule can be made, broken, transported or used without water. That is why the cells and the body need a constant supply of water.

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2. Find the amount/percentage of carbohydrates, proteins, fats and water in the following food products and compare them with each other: Roti (40 g), Burger, Cucumber, Egg, Rice (100 g), Potato, Fries (100 g), Carrot.

The following table gives the approximate amount (in grams) of carbohydrates, proteins, fats and water in each food, and their percentage of water. Values are for the usual edible portion; they may vary a little with cooking.

Food	Carbo-hydrates (g)	Proteins (g)	Fats (g)	Water (%)	Main nature
Roti (40 g)	20	4	0.5	30	Rich in starch and fibre
Burger (100 g)	30	15	13	40	Rich in fats and proteins

Food	Carbo-hydrates (g)	Proteins (g)	Fats (g)	Water (%)	Main nature
Cucumber (100 g)	3.6	0.7	0.1	96	Mostly water
Egg (100 g)	1.1	13	11	76	Rich in proteins and fats
Rice, cooked (100 g)	28	2.7	0.3	68	Rich in starch
Potato, boiled (100 g)	20	2	0.1	77	Rich in starch
Fries (100 g)	41	3.5	15	55	Rich in starch and fats
Carrot (100 g)	10	0.9	0.2	88	Mostly water, some sugar and fibre

Table: Approximate composition of common foods

Comparison:

1. Water: every food contains a large amount of water. Cucumber (96%), carrot (88%), potato (77%) and egg (76%) have the most water; fries (55%), burger (40%) and roti (30%) have less water because they are drier or fried.

2. Carbohydrates: the plant foods (rice, roti, potato, fries, burger bun) are the richest in carbohydrates, because they contain starch; cucumber, carrot and egg have very little carbohydrate.

3. Proteins: the animal foods are the best source — egg (13 g) and burger (15 g per 100 g) — and roti also gives some protein; cucumber and carrot give very little.

4. Fats: fries (15 g) and burger (13 g) and egg (11 g) are rich in fats because of frying and of animal fat; potato, carrot and cucumber have almost no fat.

Conclusion: no single food gives everything in the right amount. A balanced diet is made by mixing plant foods (carbohydrates, fibre, water) with animal foods (proteins, fats) and with fruits and vegetables (water, vitamins, minerals).

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Points to Remember

Biomolecules are the molecules produced by living organisms and used by them: carbohydrates, proteins, lipids and nucleic acids. They make up 85–90% of the dry mass of a cell; the remaining 1–1.5% is inorganic salts.

Carbohydrates are organic compounds composed of carbon, hydrogen and oxygen; they are the primary source of energy. Their three classes are monosaccharides (glucose, fructose), disaccharides (sucrose, lactose, maltose) and polysaccharides (starch, glycogen, cellulose).

Starch is the storage polysaccharide of plants, **glycogen** the animal starch stored in liver and muscles, and **cellulose** forms the plant cell wall and is not digested by humans (roughage).

Proteins are made of amino acids. About 20 types of amino acids are found in living organisms; each amino acid has an amino group, a carboxyl group and an R group, and the R group decides its properties. Amino acids are joined by peptide bonds.

Functions of proteins: growth and repair, structural support (collagen, keratin), enzymes, hormones (insulin), muscle contraction (actin, myosin), blood clotting (fibrinogen), defence (antibodies) and transport (haemoglobin).

Lipids are organic compounds that are insoluble in water but soluble in organic solvents. They are made of glycerol and fatty acids. Saturated fatty acids have only single bonds and are solid; unsaturated fatty acids have one or more double bonds and are liquid. Lipids store energy, make the cell membrane, insulate the body and help in the absorption of vitamins A, D, E and K.

Nucleic acids are made of nucleotides. Each nucleotide has a pentose sugar, a phosphate group and a nitrogenous base. **DNA** is double-stranded and contains deoxyribose, and its bases are A, T, C and G; **RNA** is single-stranded and contains ribose, with uracil in place of thymine.

DNA stores the hereditary information in the sequence of its nucleotides; it was shown as a double helix by Watson and Crick (1953). Base pairing: A with T, and G with C.

Three types of RNA: mRNA carries the message from DNA to the ribosome, tRNA brings the amino acids, and rRNA is a part of the ribosome.

Transcription is the copying of a specific sequence of DNA nucleotides in the form of mRNA; **translation** is the reading of mRNA by the ribosome to join the amino acids into a protein. DNA → RNA → protein.

Biomolecule	Building unit	Main function
Carbohydrates	Monosaccharides (glucose)	Primary source of energy; cellulose for support
Proteins	Amino acids	Growth, repair, enzymes, hormones, defence, transport
Lipids	Glycerol and fatty acids	Long-term energy store, cell membrane, insulation

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Biomolecule	Building unit	Main function
Nucleic acids	Nucleotides	Storage and transfer of hereditary information

Table: Biomolecules in one look

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d).

• Short question: 2 to 4 sentences; give the definition and one example.

• Difference question: always draw a two-column table with headings.

• Detailed answer: definition, then structure, then functions as numbered points. Draw the amino acid and nucleotide structures where asked — they carry marks.

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